Frequency Response Magnitude and Poles and Zeros

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Key Concepts

1) The frequency response at ω is proportional to the product of the distances from $e^{j\omega}$ to the zeros divided by the product of the distances from $e^{j\omega}$ to the poles.

    a) When $e^{j\omega}$ is close to a zero, $|H(e^{j\omega})|$ is small.

    b) When $e^{j\omega}$ is close to a pole, $|H(e^{j\omega})|$ is large.

2) Clusters of poles near the unit circle usually indicate that a filter has a passband for the frequencies corresponding to the angles of the pole locations.

3) Zeros near or on the unit circle are always in the stop band of the filter.

4) All zero filters (poles all at $z = 0$) have stop bands where zeros are on the circle, and passbands at frequencies where there are no zeros on the circle.

Video Screenshots

Influence of poles/zeros on $|H(e^{j\omega})|$

Stable $H(z) \Rightarrow$ ROC includes $|z|=1$

$$H(e^{j\omega}) = H(z)\Big|_{z=e^{j\omega}} = \frac{\sum_{k=0}^{M} b_k e^{-j\omega k}}{\sum_{k=0}^{N} a_k e^{-j\omega k}}$$

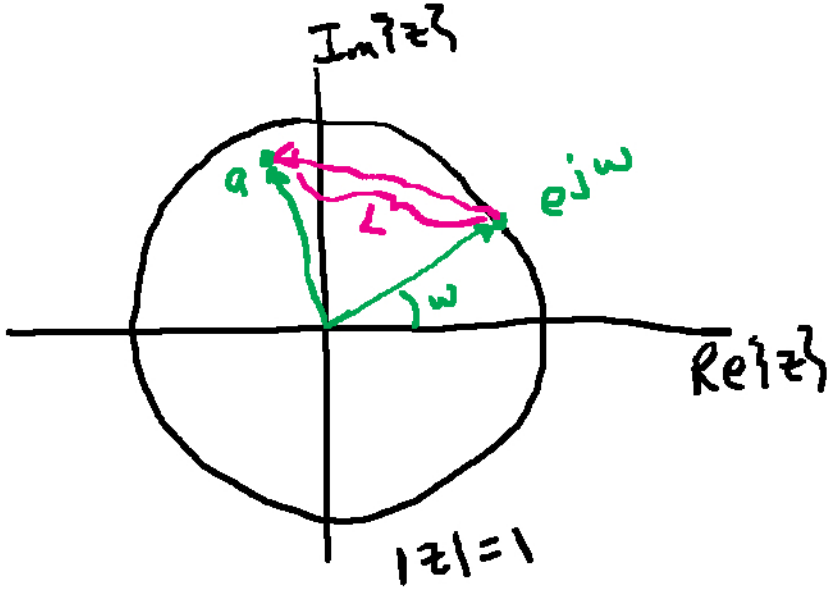
In pole/zero form

$$H(e^{j\omega}) = \frac{b_0}{a_0} \frac{\prod_{k=0}^{M}(1-c_k e^{-j\omega})}{\prod_{k=0}^{N}(1-d_k e^{-j\omega})}$$

$$|H(e^{j\omega})| = \frac{|b_0|}{|a_0|} \frac{\prod_{k=0}^{M}|1-c_k e^{-j\omega}|}{\prod_{k=0}^{N}|1-d_k e^{-j\omega}|} = \frac{|b_0|}{|a_0|} \frac{\prod_{k=0}^{M}|e^{j\omega}-c_k|}{\prod_{k=0}^{N}|e^{j\omega}-d_k|}$$

since $\dfrac{|e^{-j\omega M}|}{|e^{-j\omega N}|} = 1$

$|H(e^{j\omega})|$ depends on terms $|e^{j\omega} - a|$



$e^{j\omega} - a$ is a vector connecting $e^{j\omega}$ to a

$L = |e^{j\omega} - a|$ is the length of the vector

$$|H(e^{j\omega})| = \frac{|b_0|}{|a_0|} \frac{\prod_{k=0}^{M} |e^{j\omega} - c_k|}{\prod_{k=0}^{N} |e^{j\omega} - d_k|}$$

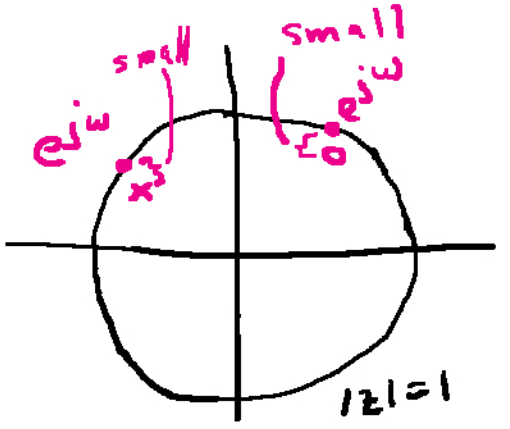
$|e^{j\omega} - c_k|$ = distance from $e^{j\omega}$ to c_k

$|e^{j\omega} - d_k|$ = distance from $e^{j\omega}$ to d_k

$$|H(e^{j\omega})| = \frac{|b_0|}{|a_0|} \frac{\prod_{k=0}^{M} \text{"distance from } e^{j\omega} \text{ to zeros"}}{\prod_{k=0}^{N} \text{"distance from } e^{j\omega} \text{ to poles"}}$$

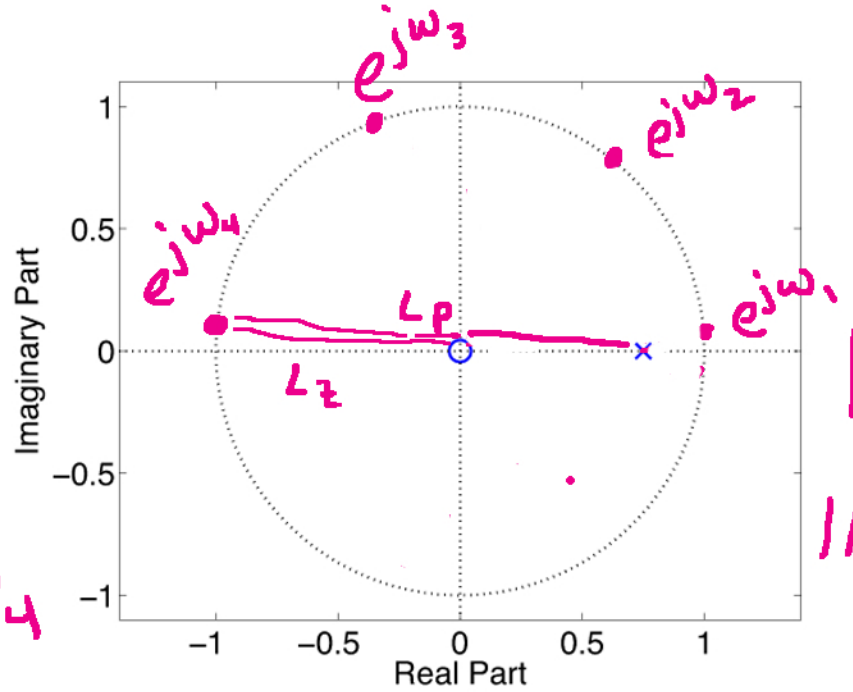
When $e^{j\omega}$ is close to a zero, $|H(e^{j\omega})|$ is "small"

When $e^{j\omega}$ is close to a pole, $|H(e^{j\omega})|$ is "large"



Example: $H(z) = \frac{1}{1-3/4z^{-1}}$

zero: $z = 0$
pole: $z = 3/4$



$L_z = 1$

$L_p = 7/4$

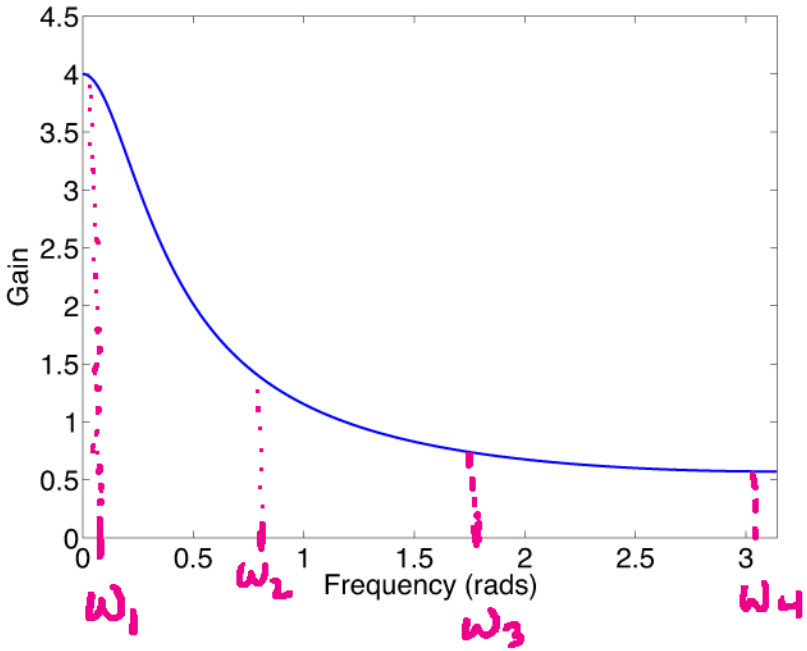
$|H(e^{j\omega_1})| \approx \frac{1}{1/4}$

$|H(e^{j\omega_2})| \approx \frac{1}{.77} \approx 1.3$

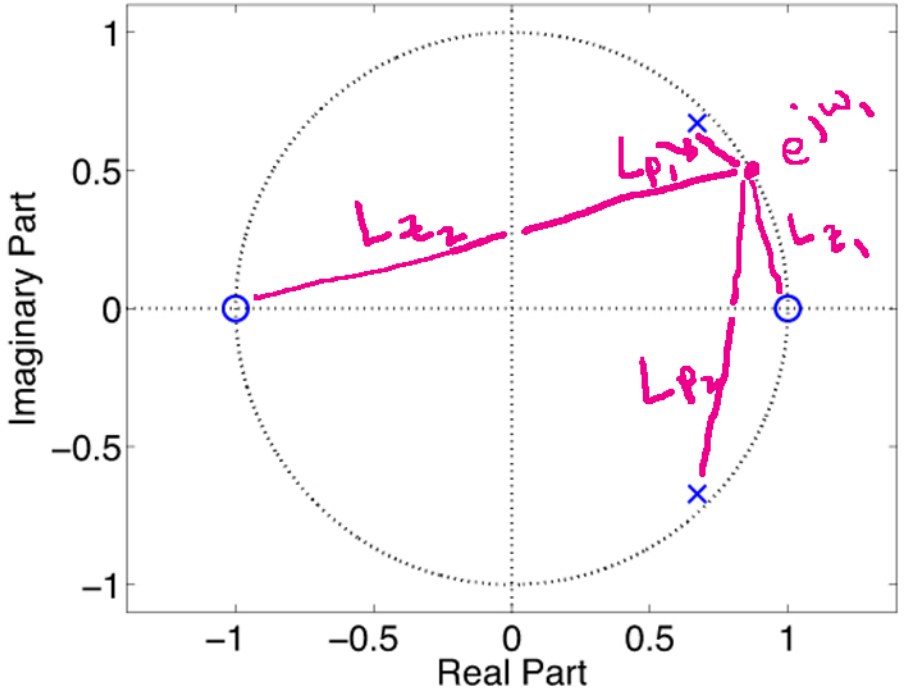
$|H(e^{j\omega_3})| \approx \frac{1}{1.25} = 0.8$

$|H(e^{j\omega_4})| \approx \frac{1}{7/4} = 4/7$

$|H(e^{j\omega})|$



Example: $H(z) = \dfrac{1 - z^{-2}}{(1 - .95e^{j\pi/4} z^{-1})(1 - .95e^{-j\pi/4} z^{-1})}$

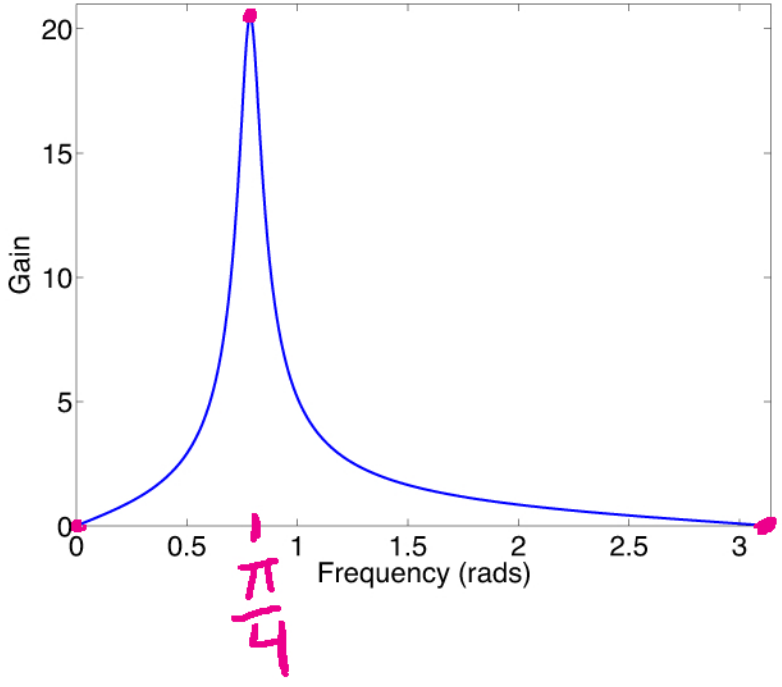


$$|H(e^{j\omega_1})| = \frac{L_{z_1} L_{z_2}}{L_{p_1} L_{p_2}}$$

$\omega_1 = 0$: $L_{z_1} = 0$

$\omega_1 = \frac{\pi}{4}$: $L_{p_1} = 0.05$

$\omega_1 = \pi$: $L_{z_2} = 0$



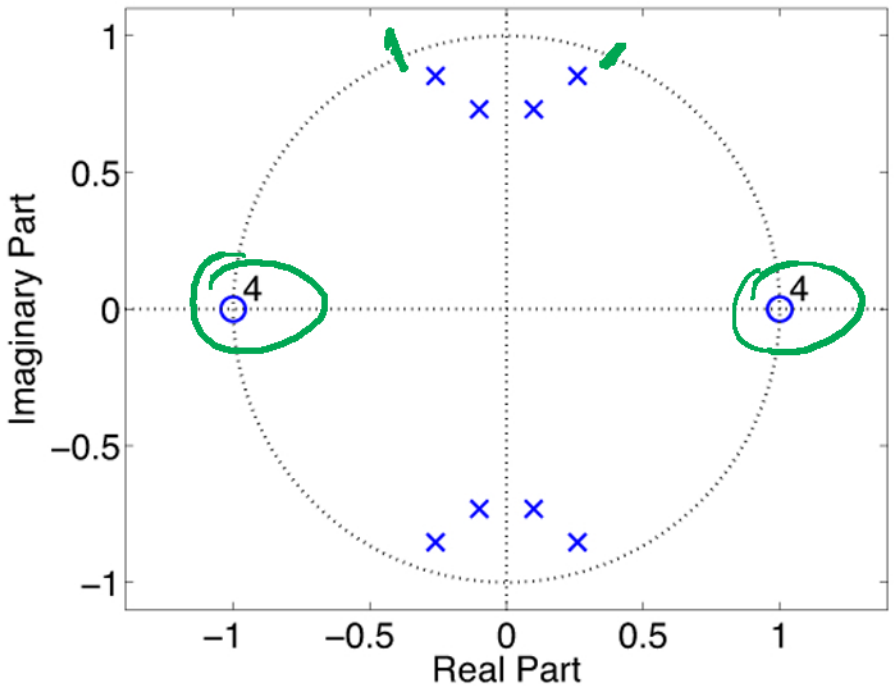
poles push magnitude response up

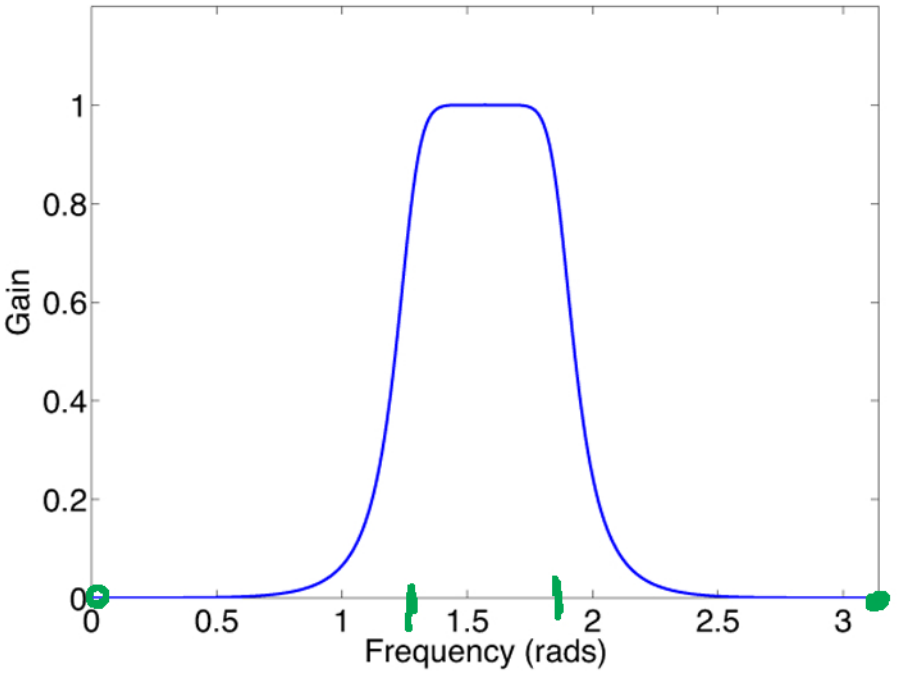
zeros pull magnitude response down

Can infer filter characteristics from pole/zero plot

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Example:

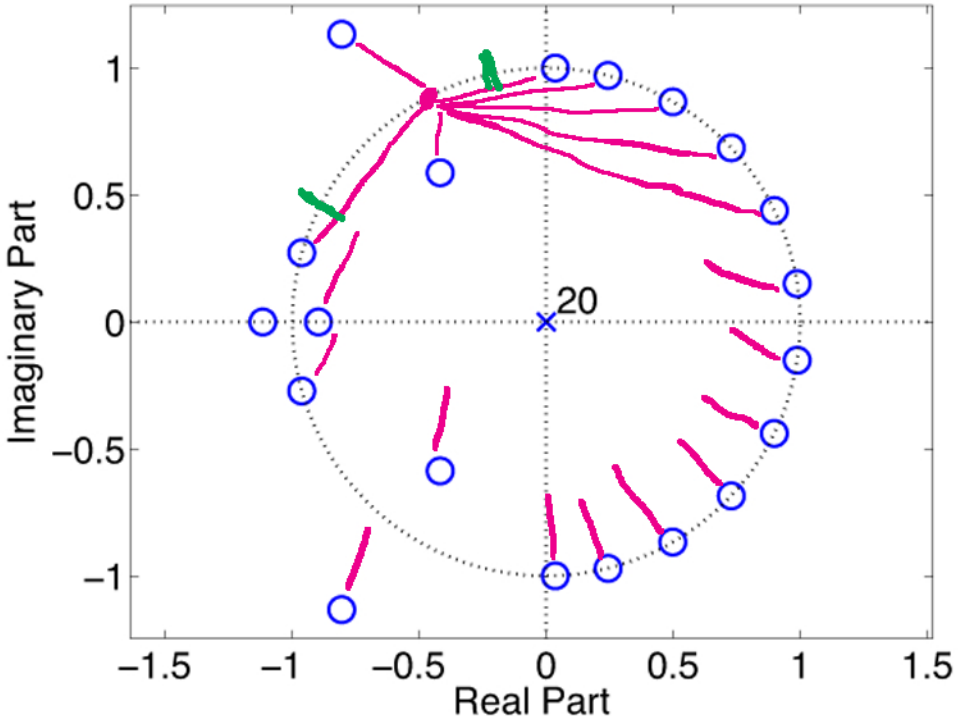


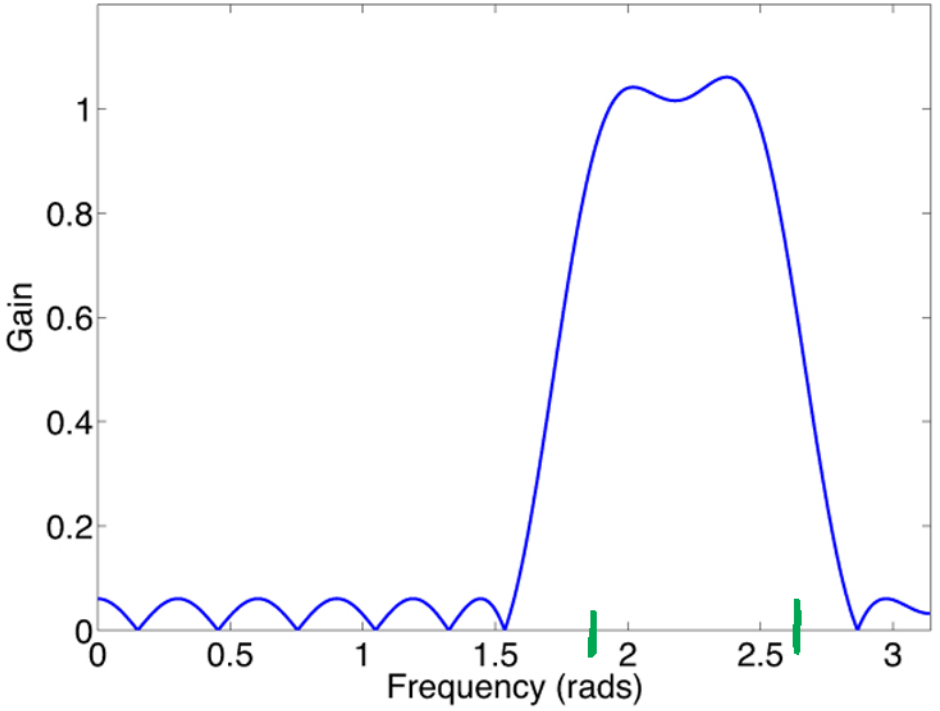


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Example:

$$|H(e^{j\omega})| = |b_0| \prod_{i=1}^{20} L_i$$





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